# The simplex package

Linguistic example environments for LaTeX

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**Abstract**

The simplex package provides three numbered example environments for use in linguistics papers—`unlabeledexample`, `shortexample`, and `example`—together with interlinear gloss macros (`\gll`, `\glll`, `\glt`, `\gln`, `\glend`) adapted from covington.sty and gloss.tex.

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## 1 Introduction

Linguistics papers routinely include numbered examples, sometimes with descriptive labels, and frequently with word-by-word interlinear glosses paired with a free translation. The simplex package provides a minimal, self-contained set of macros for both tasks.

The gloss alignment code is adapted from gloss.tex (v1.0, Marcel R. van der Goot, 1991) and covington.sty (Michael Covington). The example counter environments are original.

## 2 Requirements

simplex requires LaTeX $2_\varepsilon$ and automatically loads the xparse package (part of the standard LaTeX distribution).

## 3 Numbered example environments

All three environments share a single counter, `exampleno`, which increments automatically and is typeset as an Arabic numeral in parentheses, e.g. **(1)**, **(2)**, ...

`unlabeledexample` (*env.*) The `unlabeledexample` environment produces a numbered example with no label. It takes no arguments.

```
\begin{unlabeledexample}
The cat sat on the mat.
\end{unlabeledexample}
```

(1) The cat sat on the mat.

`shortexample` (*env.*) The `shortexample` environment produces a numbered example whose optional label appears on the *same line* as the number. The label is given as an optional argument in square brackets.

```
\begin{shortexample}[Substance-freeness of structural change]
Features changed by rule application need not be present in the rule's
structural environment.
\end{shortexample}
```

(2) Substance-freeness of Structural Change: Features changed by rule application need not be present in the rule's structural environment.

`example` (*env.*) The `example` environment is like `shortexample` except that the label appears on a *separate line* above the example content, separated by a small vertical skip. This is useful when the label is long or when the content is multi-line (such as a gloss block).

```
\begin{example}[Assibilating perfect passive participles and agents]
\begin{tabular}[H]{llllll}
metere & ‘reap’ & messus & ‘harvested’ & messor & ‘reaper’ \\
fodere & ‘dig’  & fossus & ‘dug’       & fossor & ‘digger’ \\
\end{tabular}
\end{example}
```

(3) Assibilating perfect passive participles and agents:

| | | | | | |
|---|---|---|---|---|---|
| metere | ‘reap’ | messus | ‘harvested’ | messor | ‘reaper’ |
| fodere | ‘dig’ | fossus | ‘dug’ | fossor | ‘digger’ |

### 3.1 Cross-referencing

All three environments call `\refstepcounter`, so standard `\label`/`\ref` cross-referencing works normally.

```
\begin{unlabeledexample}
```

```
The cat sat on the mat.
\label{ex:basic}
\end{unlabeledexample}

See~(\ref{ex:basic}).
```

# 4 Interlinear gloss macros

simplex provides macros for typesetting word-aligned interlinear glosses of the kind common in linguistic fieldwork and typological work. The macros accept two or three lines of space-separated tokens and stack them vertically, aligning each column.

## 4.1 Two- and three-line glosses: \gll and \glll

\gll `\gll` introduces a two-line gloss: a source-language line followed by a morpheme-gloss line. The two lines are separated by a literal newline (end of line) inside the macro argument; each line ends with `\\`. After the two gloss lines, supply a translation with `\glt` (or `\gln` if no translation is needed), and optionally close with `\glend`.

\glll `\glll` is like `\gll` but accepts three aligned lines (typically: source, morpheme gloss, and a second-tier gloss or transliteration).

```
\begin{example}[\textsc{Pre-Liquid Shortening} bled by external sandhi]
\gll clāmōr=ad cael-um
     shout=to  heaven-\textsc{acc.sg.}
\glt ‘a shout to heaven’ %(fragments of Ennius)
 Omitted---no-op here.
\glend
\end{example}
```

(4) Pre-Liquid Shortening bled by external sandhi:

| | |
|---|---|
| clāmōr=ad | cael-um |
| shout=to | heaven-acc.sg. |

‘a shout to heaven’

## 4.2 Translation and closing macros

\glt `\glt` closes the aligned gloss block and introduces a free translation. It must appear after the final gloss line even when no translation is needed (pass an empty line in that case).

\gln `\gln` is an alternative to `\glt` for use when no translation is wanted at all; it closes the gloss block without adding a newline.

\glend `\glend` is a semantic no-op provided for readability; it marks the end of a gloss block.

## 4.3 Customizing gloss fonts

Three token-list hooks control the font applied to each line:

| | |
|---|---|
| `\eachwordone` | font for the source line (default: `\rm`) |
| `\eachwordtwo` | font for the first gloss line (default: `\rm`) |
| `\eachwordthree` | font for the second gloss line (default: `\rm`) |

To italicize the source line, for example, add to your preamble:

```
\let\eachwordone=\it
```

The inter-word spacing in gloss blocks is controlled by the skip register `\glossglue`, whose default value is `5pt plus 2pt minus 1pt`.

## 5 Known limitations

- The `exampleno` counter is not reset at chapter boundaries. Users who want per-chapter numbering should add `\@addtoreset{exampleno}{chapter}` to their preamble.
- `\gll` and `\glll` rely on active end-of-line characters and may behave unexpectedly inside certain environments or macro arguments.
- Long words without spaces will not be broken across lines.

## 6 License

This package is copyright © Kyle Gorman and is distributed under the LaTeX Project Public License, version 1.3 or later (http://www.latex-project.org/lppl.txt). The gloss alignment code in `simplex.sty` was adapted from gloss.tex by Marcel R. van der Goot (1991). Additional adaptations are derived from covington.sty by Michael Covington.

## 7 Version history

**v1.0 (2026/03/02)** Initial public release.

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Numbers written in italic refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

**E**

environments:
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**G**

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